

Arachnids

PHYLUM	Arthropoda
SUBPHYLUM	Crustacea
CLASS	Arachnida
ORDERS	12
FAMILIES	661
SPECIES	103,000

This diverse group includes the familiar spiders, scorpions, and harvestmen. However, by far the most common and widespread members of the group are the ticks and mites. All arachnids have 2 main body segments and 4 pairs of legs (unlike insects, which have 3 main segments and 3 pairs of legs). Most are terrestrial animals, although about 10 percent of ticks and mites (and one species of spider) are found in freshwater.

Anatomy

An arachnid's body is divided into 2 sections. The head and thorax are fused to form a cephalothorax, which is joined to the abdomen, in some species by a narrow stalk. The cephalothorax bears 6 pairs of appendages. The first pair, known as the chelicerae, are pincer- or fang-like and used mainly for feeding. The

second pair, called the pedipalps, are leglike or enlarged with claws at the end. The other 4 pairs are walking legs, although in some species, the first pair are long and have a mainly sensory function. The abdomen may be divided into smaller segments,

or have a tail-like extension. It houses the respiratory organs: gases are exchanged via the trachea or organs known as book lungs. In spiders, the abdomen also contains silk glands.

Feeding

Most arachnids are predacious, but a few are scavengers and some mites are parasitic. Large arachnids usually rely on strength to subdue their prey but spiders, scorpions, and pseudoscorpions can also inject venom. Since arachnids have narrow mouths, they cannot ingest large pieces of food. They usually release digestive enzymes over the prey and then suck up the resulting liquid. In scorpions, small pieces of prey are partly digested in a small chamber in front of the mouth.

Some spiders rely on sight and stealth to catch their prey. Others use silk to build traps. For example, some spiders build spirals of sticky silk, called orb webs; those of some tropical species are large and strong enough to entrap small birds. Net-casting spiders make small webs, which they hold in their legs and drop over passing prey.

Reproduction

In most arachnids, the male passes a packet of sperm (or spermatophore) to the female using his pedipalps, chelicerae, or legs. The female releases the eggs, which in some scorpions hatch immediately. In some species, the young develop or hatch inside the female's body. Many arachnids show parental care by guarding their eggs.



HUNTING
Like most arachnids, this wolf spider eats live prey. It has impaled this grasshopper with its chelicerae, which each end in a hinged fang. The adjacent pedipalps are being used to manipulate the prey. Wolf spiders have acute vision and usually hunt at night.